

Solutions to JEE Advanced - 4 | JEE 2022 | Paper - 2

PHYSICS

$$1.(8) \quad (V_C)_{rms} = I_{rms} \left(\frac{1}{\omega C} \right) \quad \text{and} \quad (V_R)_{rms} = I_{rms} R$$

$$\Rightarrow \quad (V_R)_{rms} = (V_C)_{rms} (\omega RC) = (40)(200)(100)(10^{-5}) = 8V$$

$$2.(5) \quad \text{We know that after the first charging, the charge on each capacitor is } \frac{CV}{N} \text{ and } U_I = N \left(\frac{1}{2C} \left(\frac{CV}{N} \right)^2 \right) = \frac{CV^2}{2N}$$

Now, in the final steady state, let the charge on each of the original N capacitors be $\left(\frac{CV}{N} - x \right)$

Therefore, the charge on the $(N+1)^{\text{th}}$ capacitor is x

$$\text{So, applying KVL,} \quad N \left(\frac{1}{C} \left(\frac{CV}{N} - x \right) \right) = \frac{x}{C} \Rightarrow x = \frac{CV}{N+1}$$

So, in the final steady state, the charge on the original N capacitors is

$$Q_N = \frac{CV}{N} - \frac{CV}{N+1} = \frac{CV}{N(N+1)}$$

And, the charge on the $(N+1)^{\text{th}}$ capacitor is $\frac{CV}{N+1}$

$$\text{Therefore,} \quad U_F = N \left(\frac{1}{2C} \left(\frac{CV}{N(N+1)} \right)^2 \right) + \frac{1}{2C} \left(\frac{CV}{N+1} \right)^2 = \frac{CV^2}{N(N+1)} \quad \text{So, } \frac{U_F}{U_I} = \frac{1}{N+1}$$

$$3.(3) \quad \text{At steady state,} \quad \text{Heat produced} = \text{Heat lost}$$

$$\Rightarrow \quad Vi = k(T - T_0) \Rightarrow k = \frac{(50)(6)}{(120 - 20)} = 3 \text{ W/}^\circ\text{C}$$

$$4.(2) \quad \text{Electric field between the plates, } E = \frac{Q_1 - Q_2}{2\epsilon_0 A} = \frac{2Q}{\epsilon_0 A}$$

$$\text{Therefore, the potential difference, } \Delta V = Ed = \frac{2Qd}{\epsilon_0 A}$$

$$5.(2) \quad \text{Initially, the extension in the spring is } \Delta x_i = \frac{mg}{k}$$

After the electric field is switched on, it applies a downward force on the ball, and hence the equilibrium

$$\text{extension in the spring is now } \Delta x_f = \frac{mg + qE}{k}$$

$$\text{So, the amplitude of the oscillations of the ball is } \Delta x_f - \Delta x_i = \frac{qE}{k}$$

$$\text{And the angular frequency of the oscillations is } \omega = \sqrt{\frac{k}{m}}$$

$$\text{So, the maximum kinetic energy is } KE_{MAX} = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} k A^2 = \frac{q^2 E^2}{2k} = \frac{((0.01)(2000))^2}{2(100)} = 2 \text{ J}$$

6.(3) Let the instantaneous charge on the capacitor be Q and let the instantaneous current be I

Then,
$$I = + \frac{dQ}{dt}$$

From KVL:
$$-L \frac{dI}{dt} - \frac{Q}{C} = 0$$

Combining the equations, we get

$$\frac{d^2 Q}{dt^2} = -\frac{1}{LC} Q$$

Therefore,
$$Q(t) = Q_{\max} \sin(\omega t + \phi)$$

Here,
$$\omega = \frac{1}{\sqrt{LC}}$$

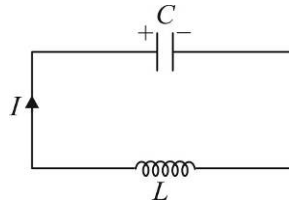
So,
$$I(t) = \frac{dQ}{dt} = \omega Q_{\max} \cos(\omega t + \phi)$$

We are given that at $t = 0$, $q = CV_0$ and $I = 0$, where $V_0 = 60$ volt

Therefore,
$$Q_{\max} = CV_0 \text{ and } \phi = \frac{\pi}{2}$$

So,
$$Q(t) = CV_0 \cos(\omega t) \text{ and } I(t) = -\omega CV_0 \sin(\omega t)$$

Therefore,
$$I_{\max} = \omega CV_0 = \frac{1}{\sqrt{LC}} CV_0 = \sqrt{\frac{C}{L}} V_0 = \sqrt{\frac{10^{-5}}{4 \times 10^{-3}}} (60) = 3 \text{ A}$$



Alternative solution

Mathematically, an LC circuit is exactly like a spring-block oscillator. The charge on the capacitor is analogous to the displacement of the block (and hence the elongation of the spring), and the current in the circuit is analogous to the velocity of the block.

Therefore, connecting a charged capacitor to an inductor, and closing the switch with the current initially zero is exactly like releasing a block from rest with the spring elongated by a certain distance.

So, just like for a spring-block oscillator, the speed of the block (or its kinetic energy) is maximum when the elongation in the spring is zero, the current in an LC circuit (or the energy stored in the inductor, $\frac{1}{2} LI^2$) is

maximum when the capacitor is completely uncharged.

Therefore, by energy conservation

$$\frac{1}{2} LI_{\max}^2 = \frac{1}{2} \frac{Q_{\max}^2}{C} \Rightarrow I_{\max} = \frac{Q_{\max}}{\sqrt{LC}} = \frac{CV_0}{\sqrt{LC}} = \sqrt{\frac{C}{L}} V_0 = 3 \text{ A}$$

7.(AD) Radius of circular path,
$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

Angular frequency of revolution,
$$\omega = \frac{qB}{m}$$

Magnetic moment,
$$M = i(\pi r^2) = \left(\frac{q\omega}{2\pi} \right) (\pi r^2) = \frac{1}{2} (q) \left(\frac{qB}{m} \right) \left(\frac{2mK}{q^2 B^2} \right) = \frac{K}{B}$$

We can check using the right hand rule that the magnetic moment is directed opposite to the magnetic field.

8.(ABD) It is clear that option (A) is correct

Let the particles carry charge Q each and let the side length of the triangle be a

$$\text{Then, initially, the field at G is } E_{G1} = 2 \left(\frac{Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{3}} \right)^2} \right) = \frac{3Q}{2\pi\epsilon_0 a^2}$$

After the particle at C is moved to a very large distance from the other particles, the electric field at G becomes

$$E_{G2} = \sqrt{3} \left(\frac{Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{3}} \right)^2} \right) = \frac{3\sqrt{3}Q}{4\pi\epsilon_0 a^2} \quad \text{So, } E_{G2} = \frac{\sqrt{3}}{2} E_{G1}$$

If (starting from the initial situation) the particle at B is moved to G, the electric field at B due to the particles at A and C is $E_{B1} = \sqrt{3} \left(\frac{Q}{4\pi\epsilon_0 a^2} \right)$ (directed along $\overline{GB}$), and the electric field at B due to the particle at G is

$$E_{B2} = \frac{Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{3}} \right)^2} = \frac{3Q}{4\pi\epsilon_0 a^2} \text{ (directed along } \overline{BG} \text{)}$$

Since $E_{B2} > E_{B1}$, the net field at B is directed along $\overline{BG}$

We can deduce easily that in the initial situation, the electric field at A is directed parallel to $\overline{CB}$

9.(AB) When dielectric slabs that cover only part of the area of the capacitor plates are introduced, the charge density on the capacitor plates is no longer uniform.

If the charge density on the lower half of the plates (i.e. in front of the points O and P) is σ_1 and $-\sigma_1$, then

$$E_O = \frac{\sigma_1}{2\epsilon_0} \quad \text{and} \quad E_P = \frac{\sigma_1}{2(K\epsilon_0)}$$

If the charge density on the upper half of the plates (i.e. in front of the points M and N) is σ_2 and $-\sigma_2$, then

$$E_M = \frac{\sigma_2}{2(2K\epsilon_0)} \quad \text{and} \quad E_N = \frac{\sigma_2}{2(K\epsilon_0)}$$

Now, the potential difference between the plates can be evaluated in two ways:

$$\Delta V = E_O \left(\frac{d}{2} \right) + E_P \left(\frac{d}{2} \right) = \frac{\sigma_1 d}{4\epsilon_0} \left(1 + \frac{1}{K} \right) \quad \text{and} \quad \Delta V = E_M \left(\frac{d}{2} \right) + E_N \left(\frac{d}{2} \right) = \frac{3\sigma_2 d}{8K\epsilon_0}$$

$$\text{Equating these expressions, we get} \quad \sigma_1 = \left(\frac{3}{2(K+1)} \right) \sigma_2$$

$$\text{Therefore,} \quad \frac{E_P}{E_O} = \frac{1}{K} ; \quad \frac{E_M}{E_O} = \frac{\sigma_2}{2K\sigma_1} = \left(\frac{K+1}{3K} \right) ; \quad \frac{E_N}{E_M} = 2$$

$$\frac{\Delta V}{E_O d} = \frac{K+1}{2K}$$

10.(AD) We know that the magnitude of the emf induced across two points on the rod that lie at distances L_1

and L_2 (such that $L_1 > L_2$) from the axis of rotation is $\varepsilon = \frac{1}{2} B \omega (L_1^2 - L_2^2)$

Therefore, the emf induced between M and Y is $\varepsilon_{MY} = \frac{1}{2} B \omega \left(\left(\frac{3L}{4} \right)^2 - 0 \right) = \frac{9}{32} B \omega L^2$

And, the emf induced between X and Y is $\varepsilon_{XY} = \frac{1}{2} B \omega \left(\left(\frac{3L}{4} \right)^2 - \left(\frac{L}{4} \right)^2 \right) = \frac{1}{4} B \omega L^2$

11.(BC) Remember that the +Z direction is along the direction $(+X) \times (+Y)$ i.e. $\hat{k} = \hat{i} \times \hat{j}$

We can visualise this by drawing a diagram showing the X-Z plane such that the wires appear points.

The field due to the wires will only have X and Z components, and its Y-component is zero

At any point on the Z-axis (the positive Z-axis or the negative Z-axis), the X-components of the field due to each wire cancel and the -Z components add up.

At the point $P(0,0,z)$, $B_P = 2 \left(\frac{\mu_0 I}{2\pi \sqrt{a^2 + z^2}} \right) \left(\frac{a}{\sqrt{a^2 + z^2}} \right) = \frac{\mu_0 I a}{2(a^2 + z^2)}$

12.(BD) The circuit is in resonance when $\omega = \frac{1}{\sqrt{LC}}$.

Therefore, we are being asked which statements are true ONLY when the circuit is in resonance and not otherwise.

(A) The charge on the capacitor (or the potential difference across it) is always behind the current in the circuit by a phase angle $\frac{\pi}{2}$. This is true for any LCR circuit, for any frequency.

(B) The rms potential difference across the capacitor is $(V_C)_{RMS} = I_{RMS} X_C = \frac{I_{RMS}}{\omega C}$

The rms induced emf across the inductor is $(V_L)_{RMS} = I_{RMS} X_L = I_{RMS} \omega L$

We can see that these quantities are equal only when $\omega = \frac{1}{\sqrt{LC}}$ and not otherwise.

(C) Since heat is only dissipated in the resistance, the rate of dissipation is $H = I^2 R$
This is always true, not only at resonance.

(D) Let the instantaneous current be $I(t) = I_0 \sin(\omega t)$

Here, I_0 is the maximum value of the current and it is a constant

Then, we know that the instantaneous potential difference across the capacitor,

$$V_C(t) = \frac{I_0}{\omega C} \sin\left(\omega t - \frac{\pi}{2}\right) = -\frac{I_0}{\omega C} \cos(\omega t)$$

The instantaneous energy stored in the capacitor, $U_C(t) = \frac{1}{2} C V_C^2 = \frac{1}{2} \frac{I_0^2}{\omega^2 C} \cos^2(\omega t)$

The instantaneous energy stored in the inductor,

$$U_L(t) = \frac{1}{2} L I^2 = \frac{1}{2} L I_0^2 \sin^2(\omega t) = \frac{1}{2} L I_0^2 (1 - \cos^2(\omega t))$$

Hence, the total energy stored in both $U(t) = U_C(t) + U_L(t) = \frac{1}{2} L I_0^2 + \frac{1}{2} I_0^2 \left(\frac{1}{\omega^2 C} - L \right) \cos^2(\omega t)$

We can see that $U(t)$ remains constant with time only if $\frac{1}{\omega^2 C} - L = 0$, i.e. at resonance.

13.(14) Velocity after $t = 1\text{ s}$ is v

$$mv = mv_0 - \mu(mg + qE)t$$

$$= 9 - 0.1(18 + 2) \times 1 = 7 \text{ kg } \frac{\text{m}}{\text{s}}$$

$$R = \frac{mv}{qB} = \frac{7}{1 \times 0.5} = 14 \text{ m}$$

14.(2) Let the instantaneous currents in the coils be I_1 and I_2

Since the coils are in parallel, the induced EMF across both of them is always equal

Therefore,
$$L_1 \frac{dI_1}{dt} = L_2 \frac{dI_2}{dt} \Rightarrow L_1 dI_1 = L_2 dI_2$$

Integrating,
$$\int_0^{I_1} L_1 dI_1 = \int_0^{I_2} L_2 dI_2$$

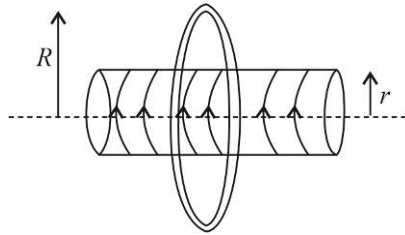
Therefore, at any instant, $L_1 I_1 = L_2 I_2$

15.(6) $\phi_{\text{initial}} = (\mu_0 n i_0) \pi r^2$

$$\phi_{\text{final}} = \left(\mu_0 \frac{n i_0}{3} \right) \pi r^2 + L i$$

$$i = \frac{2\mu_0 n \pi r^2 i_0}{3L}$$

$$= \frac{2}{3} r^2 i_0 = \frac{2}{3} \times \frac{1}{4} \times i_0 = \frac{i_0}{6}$$



16.(0.31) Let the charge per unit area be σ

Then,
$$\sigma = \frac{Q}{\pi R^2 - \pi \left(\frac{R}{2} \right)^2} = \frac{4Q}{3\pi R^2}$$

Consider a ring element of radius r and thickness dr

Charge on the element,
$$dq = \sigma(2\pi r dr)$$

Equivalent current due to rotation of the element,
$$dI = dq \left(\frac{\omega}{2\pi} \right) = \sigma \omega r dr$$

Magnetic moment of the element,
$$dm = dI(\pi r^2) = \pi \sigma \omega r^3 dr$$

Since the magnetic moment of each element will be in the same direction (parallel to the axis of rotation), the net magnetic moment,

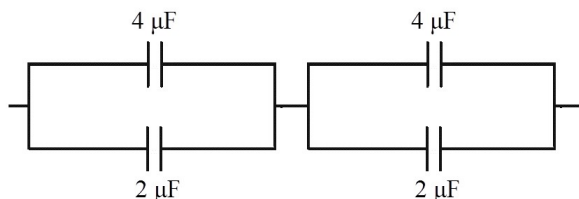
$$m = \int dm = \pi \sigma \omega \int_{R/2}^R r^3 dr = \frac{15}{64} \pi \sigma \omega R^4 \quad \text{So, } m = \frac{15}{64} \pi \omega R^4 \left(\frac{4Q}{3\pi R^2} \right) = \frac{5}{16} Q R^2 \omega$$

17.(3) In general, connecting capacitors in parallel leads to a higher net capacitance, and connecting them in series leads to a lower net capacitance

But connecting them in series also ensures that the potential difference across the combination can be made higher without damaging the capacitors

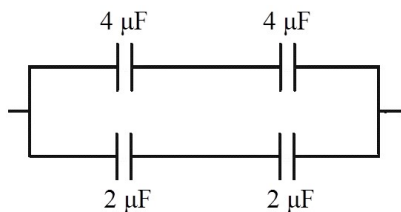
So, the capacitance is maximum when all the capacitors are in parallel, and hence let us begin with this combination, and move towards the combination with all of them in series

- (i) All in parallel; net capacitance = $4 + 4 + 2 + 2 = 12 \mu\text{F}$; but p.d. across each is 100 V, so they all get damaged
- (ii) Now, the next possibility is three of the capacitors in parallel and the fourth in series with their combination. But in this situation, the p.d. across the two series components will be unequal, and hence one of these p.d.s will exceed 100 V, so either all of the capacitors in parallel with each other get damaged, or the fourth one gets damaged
- (iii) The same argument about the p.d. being unequal works with any combination where there are two components (themselves composed of a parallel combination of some capacitors) in series, **unless** the capacitance of the two components is exactly equal, so the p.d. across each is exactly 100 V. This is possible with keeping two parallel combinations of $4 \mu\text{F}$ and $2 \mu\text{F}$ in series



This combination gives a net capacitance $3 \mu\text{F}$

- (iv) We can be sure that the above combination gives the maximum possible capacitance because a combination with three series components will give a lower capacitance
We could have done it in another way: by starting with all in parallel, and then forming parallel combinations of capacitors in series. This would give us another solution, with the same net capacitance, i.e. $3 \mu\text{F}$:



18.(280) Average power dissipated in the coil,

$$P = \frac{V^2 R}{2Z^2} = \frac{V^2 R}{2(R^2 + (\omega L)^2)} = \frac{(300)^2 (9)}{2(9^2 + ((300)(0.04))^2)} = 1800 \text{ W}$$

Total heat needed to raise the temperature of the coil, the water and the vessel to 60°C ,

$$H = (2(4200) + 3000 + 1200)(60 - 20) = 504 \text{ kJ}$$

Therefore, the time taken, $t = \frac{H}{P} = \frac{504 \times 10^3}{1800} = 280 \text{ s}$

CHEMISTRY

1.(4) $r = k[\text{NO}]^a [\text{H}_2]^b$

$$-\frac{dp}{dt} = k[\text{NO}]^a [\text{H}_2]^b$$

$$1.5 = k(372)^a [\text{H}_2]^b \quad \dots(1)$$

$$0.25 = k(152)^a [\text{H}_2]^b \quad \dots(2)$$

$$\Rightarrow 6 = (2.45)^a \Rightarrow a = 2$$

$$1.6 = k[\text{NO}]^2 (288)^b \quad \dots(3)$$

$$0.8 = k[\text{NO}]^2 (144)^b \quad \dots(2)$$

$$2 = (2)^b \Rightarrow b = 1$$

$$(a + 2b) = 2 + 2 = 4$$

2.(2) $d_{\text{exp.}} = 1.6075 \text{ g / cm}^3$

$$d_{\text{theo.}} = \frac{4 \times 31.25 \times 1.67 \times 10^{-24}}{125 \times 10^{-24}} = 1.67 \text{ g / cm}^3$$

$$\text{No. of moles of metal atoms missing per ml of crystal} = \frac{0.0625}{31.25} = 2 \times 10^{-3}$$

$$\text{No. of moles of metal atoms missing per litre of crystal} = 2 \times 10^{-3} \times 1000 = 2$$

3.(6) $0.29 = i \times 1.86 \times \frac{1.04 \times 1000}{267 \times 100}$

$$i = 4$$

Complex is $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

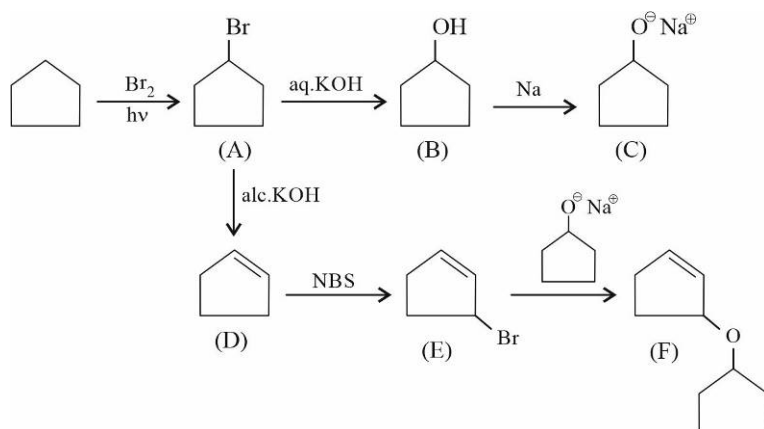
4.(3) $(\text{eq})_{\text{Cl}_2} = (\text{eq})_{\text{M}}$

$$\frac{9.08}{22.4} \times 2 = \frac{52.8}{200} \times x \quad (x \text{ is valency of metal})$$

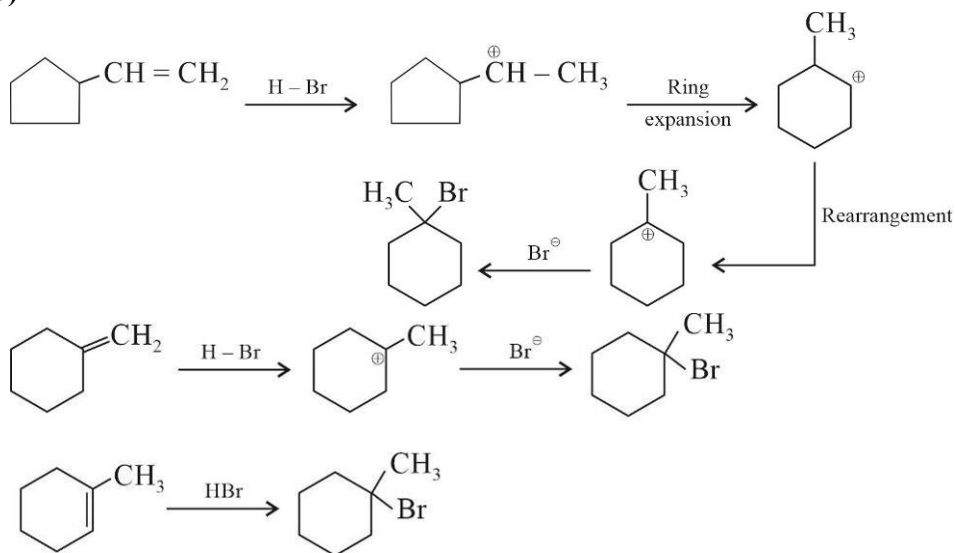
$$x = 3$$

5.(4) (ii), (iii), (v), (viii) give +ve iodoform test.

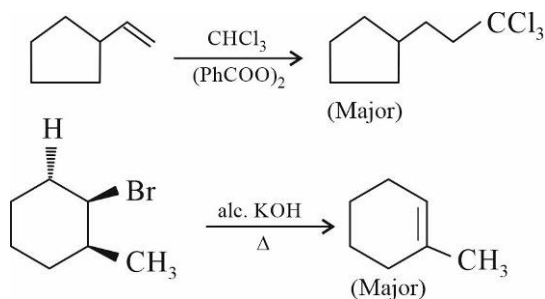
6.(3)



7.(ABC)



8.(AD)



9.(ACD) Refer to solid state theory

10.(ABC) $k = Ae^{-E_a/RT}$

$$\frac{dk}{dT} = Ae^{-E_a/RT} \frac{E_a}{RT^2} = k \cdot \frac{E_a}{RT^2}$$

$$\frac{1}{k} \frac{dk}{dT} = \frac{E_a}{RT^2} = \frac{1.25 \times 10^6}{T^3}$$

$$\frac{E_a}{R} = \frac{1.25 \times 10^6}{T}$$

$$\text{At } 250\text{K}, E_a = \frac{2 \times 1.25 \times 10^6}{250} = 10000 \text{ Cal/mol} = 10 \text{ kcal/mol}$$

$$\text{At } 2000\text{K}, E_a = \frac{2 \times 1.25 \times 10^6}{2000} = 1250 \text{ cal/mol} = 1.25 \text{ kcal/mol}$$

$$\frac{d(\ln k)}{dT} = \frac{E_a}{R} \left(\frac{1}{T^2} \right)$$

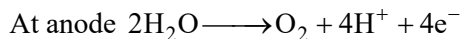
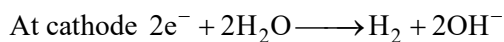
$$\text{At } 1000\text{K}, \frac{d(\ln k)}{dT} = \frac{1.25 \times 1000}{2} \times \frac{1}{10^6} = 6.25 \times 10^{-4}$$

11.(ABC) From the order of coagulation power, it is clear that it is positively charged colloidal solution

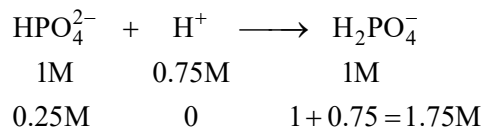
12.(BC) H_2PO_4^- HPO_4^{2-}

Initially 1M 1M

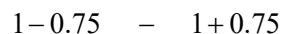
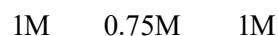
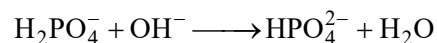
$$\text{No. of Faradays of electricity} = \frac{1.25 \times 965 \times 60}{96500} = 0.75$$



So at anode

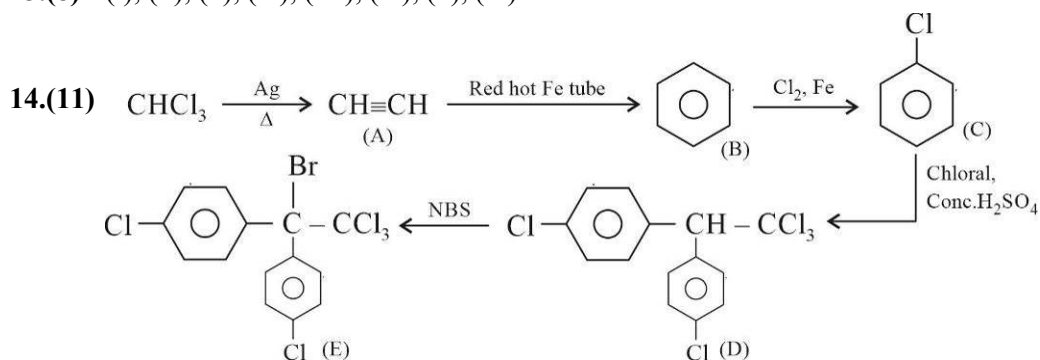


$$\text{p}^{\text{H}} = 2.15 + \log \frac{0.25}{1.75} = 2.15 - 0.85 = 1.3$$

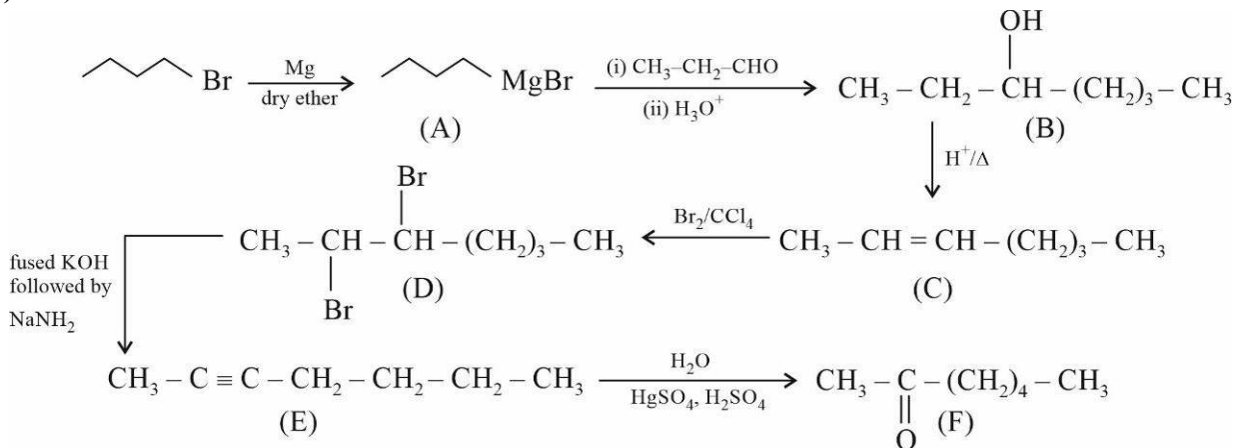


$$\text{p}^{\text{H}} \text{ at cathode: } 2.15 + \log \frac{1.75}{0.25} = 2.15 + 0.85 = 3$$

13.(8) (i), (ii), (v), (vi), (vii), (ix), (x), (xi)



15.(114)



Molar mass = 114 g/mol

$$16.(108) (\lambda_{eq}^{\infty})Ba_3(PO_4)_2 = 6(160 + 140 - 100) = 1200 \text{ ohm}^{-1}\text{cm}^2\text{eq}^{-1}$$

$$(k)Ba_3(PO_4)_2 = 1.2 \times 10^{-5} \text{ ohm}^{-1}\text{cm}^{-1}$$

$$S = 1.2 \times 10^{-5} \times \frac{1000}{1200} = 10^{-5}$$

$$(K_{sp})Ba_3(PO_4)_2 = 108 \times 10^{-25}$$

$$17.(4.80) \Delta T_f = K_f \cdot m$$

$$0.4 = 1.86 \times m \Rightarrow m = \frac{0.4}{1.86}$$

$$\pi = \frac{0.4}{1.86} \times 0.08 \times 279 = 4.8 \text{ atm}$$

$$18.(12.75) \text{ Total number of } NH_3 \text{ molecules} = \frac{10^{16} \times 10^4}{(10^{-3} / 18)} = 18 \times 10^{23}$$

$$\text{No. of } NH_3 \text{ molecules adsorbed per g of charcoal} = \frac{18 \times 10^{23}}{4}$$

$$\text{No. of moles of } NH_3 \text{ adsorbed per g of charcoal} = \frac{18 \times 10^{23}}{24 \times 10^{23}} = \frac{3}{4}$$

$$\text{Mass of } NH_3 \text{ adsorbed per g of charcoal} = \frac{3}{4} \times 17 = 12.75 \text{ g}$$

MATHEMATICS

1.(4) We have $F(x) + F\left(x + \frac{1}{2}\right) = 3$ (i)

Replace x by $x + \frac{1}{2}$ in (i), we get $F\left(x + \frac{1}{2}\right) + F(x+1) = 3$ (ii)

$\therefore$ From (i) and (2), we get $F(x) = F(x+1)$ (iii)

$\Rightarrow F(x)$ is periodic function.

Now consider $I = \int_0^{1500} F(x) dx = \int_0^1 F(x) dx + \int_1^{1500} F(x) dx$

(Using property of periodic function)

Put $x = y + \frac{1}{2}$ 2nd integral, we get

$$I = 1500 \left[\int_0^{1/2} F(x) dx + \int_0^{1/2} F\left(y + \frac{1}{2}\right) dy \right] = 1500 \int_0^{1/2} \left(F(x) + F\left(x + \frac{1}{2}\right) \right) dx = 1500 \int_0^{1/2} 3 dx \quad (\text{Using (i)})$$

Hence $I = 1500(3) \left(\frac{1}{2} \right) = 750 \times 3 = 2250$

2.(4) $I = \int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx = \int_0^{2\pi} \frac{(2\pi - x) \sin^8 x}{\sin^8 x + \cos^8 x} dx$

$$\Rightarrow I = \int_0^{2\pi} \frac{\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx = 4 \int_0^{\pi/2} \frac{\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx = 4 \int_0^{\pi/2} \frac{\pi \cos^8 x}{\cos^8 x + \sin^8 x} dx$$

$$\therefore 2I = 4\pi \int_0^{\pi/2} \frac{\sin^8 x + \cos^8 x}{\cos^8 x + \sin^8 x} dx = 2\pi^2$$

$\therefore I = \pi^2$. Hence $k = 4$

3.(2) Given $\frac{dy}{dx} - \frac{1}{x}y = \left(x - \frac{2}{x}\right)$

I.F. = $e^{-\int \frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}$

Now general solution is given by $\frac{y}{x} = \int \left(x - \frac{2}{x}\right) \frac{1}{x} dx$

$$\Rightarrow \frac{y}{x} = x + \frac{2}{x} + C$$

As $y(1) = 1 \Rightarrow C = -2$

$$\therefore \frac{y}{x} = x + \frac{2}{x} - 2 \Rightarrow y = x^2 - 2x + 2$$

Hence $y(2) = (2)^2 - 2(2) + 2 = 2$

$$4.(6) \quad 6 \int_1^x f(t) dt = 3xf'(x) - x^3$$

$$\Rightarrow 6f(x) = 3f(x) + 3xf'(x) - 3x^2 \Rightarrow xf'(x) - f(x) = x^2$$

$$\Rightarrow x \frac{dy}{dx} - y = x^2 \Rightarrow \frac{xdy - ydx}{x^2} = dx \Rightarrow \int \frac{xdy - ydx}{x^2} = \int dx \Rightarrow \frac{y}{x} = x + c$$

$$\text{Given } f(1) = 2 \Rightarrow c = 1 \Rightarrow y = x^2 + x$$

$$5.(3) \quad \text{Area of quadrilateral } OABC = \Delta OAC + \Delta ABC$$

$$= \frac{1}{2} |\vec{OA} \times \vec{AC}| + \frac{1}{2} |\vec{AB} \times \vec{BC}| = \frac{1}{2} |\vec{a} \times (\vec{b} - \vec{a})| + \frac{1}{2} |(2\vec{a} + 10\vec{b} - \vec{a}) \times (\vec{b} - 2\vec{a} - 10\vec{b})|$$

$$= \frac{1}{2} |\vec{a} \times \vec{b}| + \frac{1}{2} |\vec{a} \times 10\vec{b}| = 6 |\vec{a} \times \vec{b}|$$

$$|\vec{a} \times \vec{b}| = m ; \lambda = 2\lambda m \Rightarrow 6 |\vec{a} \times \vec{b}| = 2\lambda |\vec{a} \times \vec{b}| \Rightarrow \lambda = 3$$

$$6.(1) \quad \hat{i} \times [(\vec{a} - \hat{j}) \times \hat{i}] = (\hat{i} \cdot \hat{i})(\vec{a} - \hat{j}) - (\hat{i} \cdot (\vec{a} - \hat{j}))\hat{i} = \vec{a} - \hat{j} - (\hat{i} \cdot \vec{a})\hat{i}$$

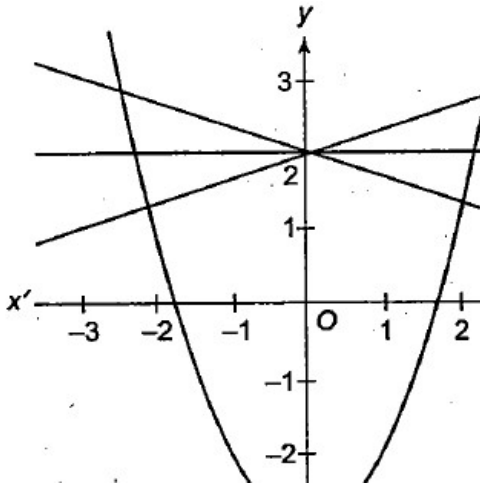
$$\therefore \vec{a} - \hat{j} - (\hat{i} \cdot \vec{a})\hat{i} + \vec{a} - \hat{k} + (\hat{j} \cdot \vec{a})\hat{j} + \vec{a} - \hat{i} - (\hat{k} \cdot \vec{a})\hat{k} = 0$$

$$3\vec{a} - (\hat{i} + \hat{j} + \hat{k}) - \vec{a} = 0$$

$$\vec{a} = \frac{1}{2}(\hat{i} + \hat{j} + \hat{k}) = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore x = y = z = \frac{1}{2} \Rightarrow 8(x^3 - xy + zx) = 8(x^3 - x^2 + x^2) = 8 \times \frac{1}{8} = 1$$

7.(BC)



Line $y = kx + 2$ passes through fixed point $(0, 2)$ for different value of k .

Also, it is obvious that minimum $A(k)$ occurs when $k = 0$, as when line is rotated from this position about point $(0, 2)$ the increased part of area is more than the decreased part of area.

$$\therefore \text{Minimum area} = 2 \int_0^5 (2 - (x^2 - 3)) dx$$

$$= 2 \int_0^5 (5 - x^2) dx = 2 \left[5x - \frac{x^3}{3} \right]_0^5 = 2 \left[5\sqrt{5} - \frac{5\sqrt{5}}{3} \right] = \frac{20\sqrt{5}}{3} \text{ sq.units}$$

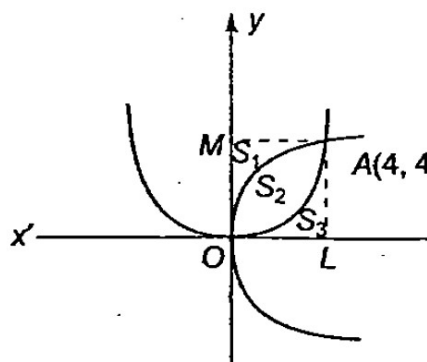
8.(ACD) $y^2 = 4x$ and $x^2 = 4y$ meet at $O(0, 0)$ and $A(4, 4)$

$$\text{Now } S_3 = \int_0^4 \frac{x^2}{4} dx = \frac{1}{4} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{12} [64 - 0] = \frac{16}{3}$$

$$S_2 = \int_0^4 2\sqrt{x} dx - S_3 = 2 \left[\frac{x^{3/2}}{3/2} \right]_0^4 = \frac{16}{3}$$

$$= \frac{4}{3} [8 - 0] - \frac{16}{3} = \frac{16}{3}$$

and $S_1 = 4 \times 4 - (S_2 + S_3) = 16 - \left(\frac{16}{3} + \frac{16}{3} \right) = \frac{16}{3}$ Hence, $S_1 : S_2 : S_3 = 1 : 1 : 1$



9.(AB) $x = \sin \left(\frac{dy}{dx} - 2y \right) \Rightarrow \frac{dy}{dx} - 2y = \sin^{-1} x$; $x - 2y = \log \left(\frac{dy}{dx} \right) \Rightarrow \frac{dy}{dx} = e^{x-2y}$

10.(ACD)

(A) Since $\vec{n} = \lambda(\vec{a} \times \vec{b})$, $\vec{n} \cdot \vec{c} = 0$ $\therefore [\vec{a} \vec{b} \vec{c}] = 0$

(B) $\cos^2 30^\circ + \cos^2 45^\circ + \cos^2 \gamma = 1$ $\therefore \sin^2 \gamma = \frac{3}{4} + \frac{1}{2} = \frac{5}{4}$ which is not possible

(C) Obvious

(D) $\vec{AB} \times \vec{BC} = (\vec{OB} - \vec{OA}) \times (\vec{OC} - \vec{OB}) = \vec{OB} \times \vec{OC} - \vec{OA} \times \vec{OC} + \vec{OA} \times \vec{OB}$
 $= \vec{OA} \times \vec{OB} + \vec{OB} \times \vec{OC} + \vec{OC} \times \vec{OA}$

i.e., $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is perpendicular to the plane ABC.

Now, $(\vec{a} + \vec{b} + \vec{c}) \times (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$

$$= ((\vec{a} + \vec{b} + \vec{c}) \vec{b} - ((\vec{a} + \vec{b} + \vec{c}) \vec{a}) \vec{b} + ((\vec{a} + \vec{b} + \vec{c}) \vec{c}) \vec{b} - ((\vec{a} + \vec{b} + \vec{c}) \vec{b}) \vec{c} \\ + ((\vec{a} + \vec{b} + \vec{c}) \vec{a}) \vec{c} - ((\vec{a} + \vec{b} + \vec{c}) \vec{a}) \vec{a} \\ = (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{b} - \vec{c}) \vec{a} + (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{c} - \vec{a}) \vec{b} + ((\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} - \vec{b})) \vec{c} \\ = (\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} + \vec{b}^2 - \vec{c}^2) \vec{a} + (\vec{b} \cdot \vec{c} - \vec{b} \cdot \vec{a} + \vec{c}^2 - \vec{a}^2) \vec{b} + (\vec{a}^2 - \vec{b}^2 + \vec{c} \cdot \vec{a} - \vec{c} \cdot \vec{b}) \vec{c} = 0$$

Hence, the statement is true

11.(AD) $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = -7\vec{b}$ and this is orthogonal to $\vec{a}$

$\vec{a} + \vec{b} + \vec{c} = 4\hat{i} + 3\hat{j} + \hat{k}$ is also orthogonal to $\vec{b}$

12.(AC)

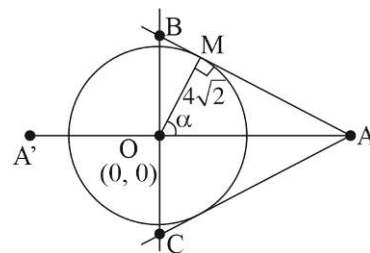
$OM = 4\sqrt{2}$

$OA = 4\sqrt{2} \sec \alpha$

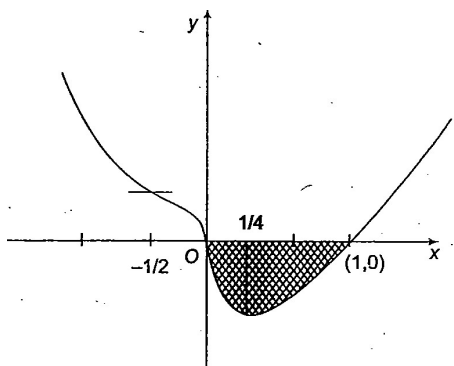
$BC = 2OB = 8\sqrt{2} \csc \alpha$

$\therefore \text{Area of } \triangle ABC = \frac{1}{2} OA \times BC = \frac{4\sqrt{2} \cdot 8\sqrt{2}}{2} \sec \alpha \csc \alpha = \frac{64}{\sin^2 \alpha}$

For min. area $\alpha = \frac{\pi}{4}$ $\therefore$ Point A is (8, 0) symmetrically $A'(-8, 0)$

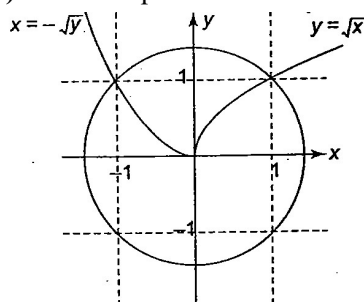


13.(0.32) Graph of $f(x)$ is as



$$A = \int_0^1 (x^{4/3} - x^{1/3}) dx = \left[\frac{3}{7} x^{7/3} - \frac{3}{4} x^{4/3} \right]_0^1 = \left| \frac{3}{7} - \frac{3}{4} \right| = \left| \frac{4-7}{28} \right| = \frac{9}{28} \Rightarrow A = \frac{9}{28} = 0.32$$

14.(0.50) Required area = area of one quadrant of the circle = $\pi/2$



15.(-0.50) $\frac{dy}{dx} - y = 1 - e^{-x}$

$$P = -1 \quad Q = 1 - e^{-x}$$

$$\text{I.F.} = e^{\int P dx} = e^{\int -1 dx} = e^{-x}$$

$$\therefore y \cdot e^{-x} = \int e^{-x} (1 - e^{-x}) dx + C$$

$$y e^{-x} = -e^{-x} + \frac{1}{2} e^{-2x} + C$$

$$y = -1 + \frac{1}{2} e^{-x} + C e^x$$

$$\therefore x = 0 \quad y = y_0$$

$$\text{So} \quad C = y_0 + \frac{1}{2}$$

$$y = -1 + \frac{1}{2} e^{-x} + (y_0 + 1/2) e^x$$

$$x \rightarrow \infty \quad y \rightarrow \text{finite value so } y_0 + \frac{1}{2} = 0$$

$$y_0 = -\frac{1}{2}$$

16.(1.50) $y = u^m$

$$\frac{dy}{dx} = mu^{m-1} \frac{du}{dx}$$

The given differential equation becomes

$$2x^4 \cdot u^m \cdot mu^{m-1} \frac{du}{dx} + u^{4m} = 4x^5$$

$$\Rightarrow \frac{du}{dx} = \frac{4x^5 - u^{4m}}{2mx^4 u^{2m-1}}$$

For homogeneous equation degree should be same in numerator & denominator so,

$$6 = 4m = 4 + 2m - 1 \Rightarrow m = \frac{3}{2}$$

17.(4) $y(1+xy)dx = xdy$ or $\frac{dy}{dx} = \frac{y(1+xy)}{x}$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} + y^2 \text{ or } \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \frac{1}{x} = 1$$

Put $-\frac{1}{y} = t$

18.(−4) Let $I = \int \frac{(x^2-1)dx}{x^3\sqrt{2x^4-2x^2+1}}$ [dividing numerator and denominator by x^5]

$$= \int \frac{\left(\frac{1}{x^3} - \frac{1}{x^5}\right)dx}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}} \text{ Put } 2 - \frac{2}{x^2} + \frac{1}{x^4} = t \Rightarrow \left(\frac{4}{x^3} - \frac{4}{x^5}\right)dx = dt \therefore I = \frac{1}{4} \int \frac{dt}{\sqrt{t}} = \frac{1}{4} \cdot \frac{t^{1/2}}{1/2} + c = \frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c$$